Synthesis to Physical Design of 8-bit Counter

EEE 4122 / EEE 442 (B): VLSI Design Laboratory

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Chapter 1

Introduction

In this project, we have designed and implemented an 8-bit counter starting from the RTL level up to the final GDSII layout. The main goal of this work is to understand the complete flow of VLSI design – how a digital circuit written in Verilog is finally converted into a physical layout ready for fabrication.

The counter is a simple and commonly used digital circuit that increases its value step by step with every clock pulse. Even though it is a small design, it helps to learn the whole design process clearly. Starting from the behavioral Verilog code, we performed synthesis to get the gate-level netlist and verified its timing, area, and power. After that, we moved to the physical design stage where floorplanning, placement, clock tree synthesis, and routing were done using Cadence Innovus.

Finally, the layout was checked for geometry and connectivity errors to make sure the design was clean. All the verification reports showed zero violations, which means the design is correct and fabrication-ready. This project helped us gain hands-on experience in the full RTL to GDSII flow and a better understanding of digital IC implementation in 45nm technology.

Chapter 2

RTL Design

2.1 Verilog RTL Code

The 8-bit counter increments its count value at every positive edge of the clock. An asynchronous reset sets the counter to zero.

Listing 2.1: 8-bit Counter RTL Code

```
module cntr8bit(input clk, reset,
                output reg [7:0] count);
always @(posedge clk or negedge reset)
begin
    if (reset==0)
        count <= 8'b00000000;
    else
        count <= count + 1;
end
endmodule
```

2.2 Testbench Code

The testbench generates the clock and reset signals to verify the functionality of the counter.

Listing 2.2: Testbench for 8-bit Counter

```
module cntr8bit_tb;

    reg clk, reset;
    wire [7:0] count;

    cntr8bit DUT (
        .clk(clk),
        .reset(reset),
        .count(count)
    );

    always #5 clk = ~clk;
```

```
    initial begin
        clk = 0;
        reset = 1;

        @(posedge clk);
        reset = 0;
        @(posedge clk);
        reset = 1;
        repeat (10) @(posedge clk);
    end

    initial begin
        $dumpfile("dump.vcd");
        $dumpvars;
        #100; $finish;
    end
endmodule
```

2.3 Simulation Waveform

The waveform below shows the clock, reset, and count signals during simulation.

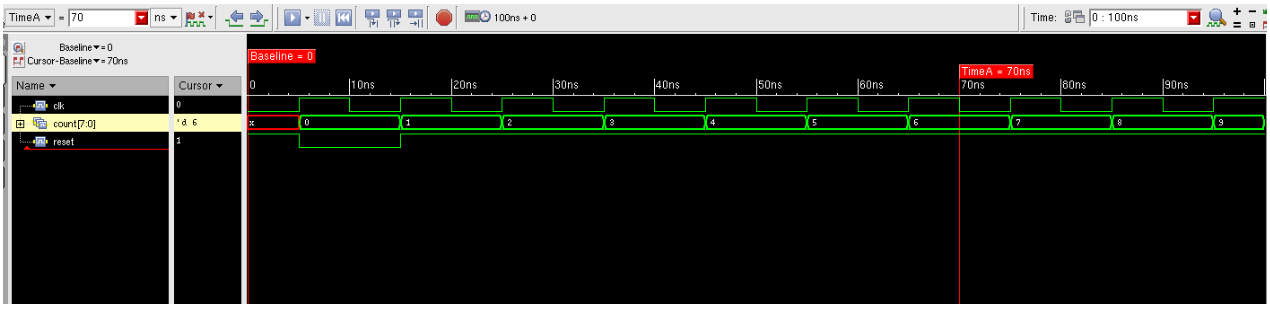

Figure 2.1: Simulation Waveform of 8-bit Counter

Chapter 3

Synthesis

The design is synthesized using the 45nm standard cell library. The RTL is converted into a gate-level netlist that meets timing and area constraints.

3.1 Synthesis Result

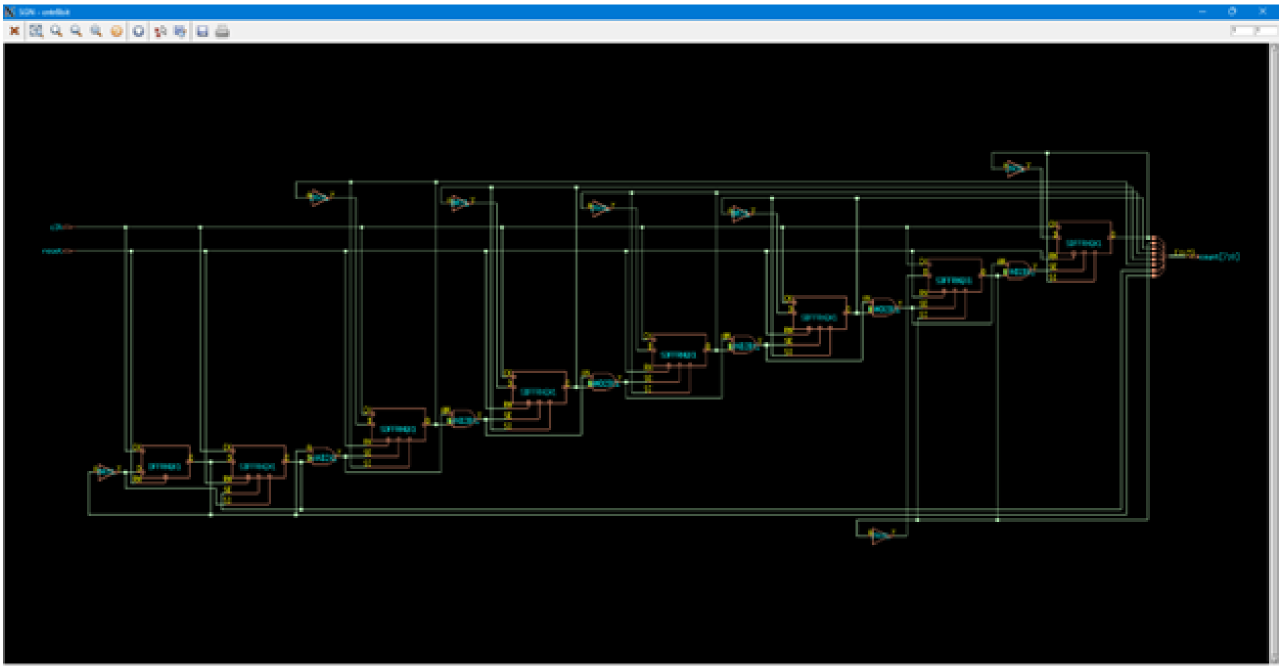

Figure 3.1: Gate-level Schematic after Synthesis

3.2 Timing, Area, and Power Report

- **Worst Negative Slack (WNS):** 7657 ps
- **Total Cell Area:** 775
- **Total Dynamic Power:** 10378.253 nW
- **Leakage Power:** 23.165 nW

Chapter 4

Physical Design

The physical design process includes floorplanning, placement, clock tree synthesis (CTS), routing, and final layout generation. All stages were completed using Cadence Innovus.

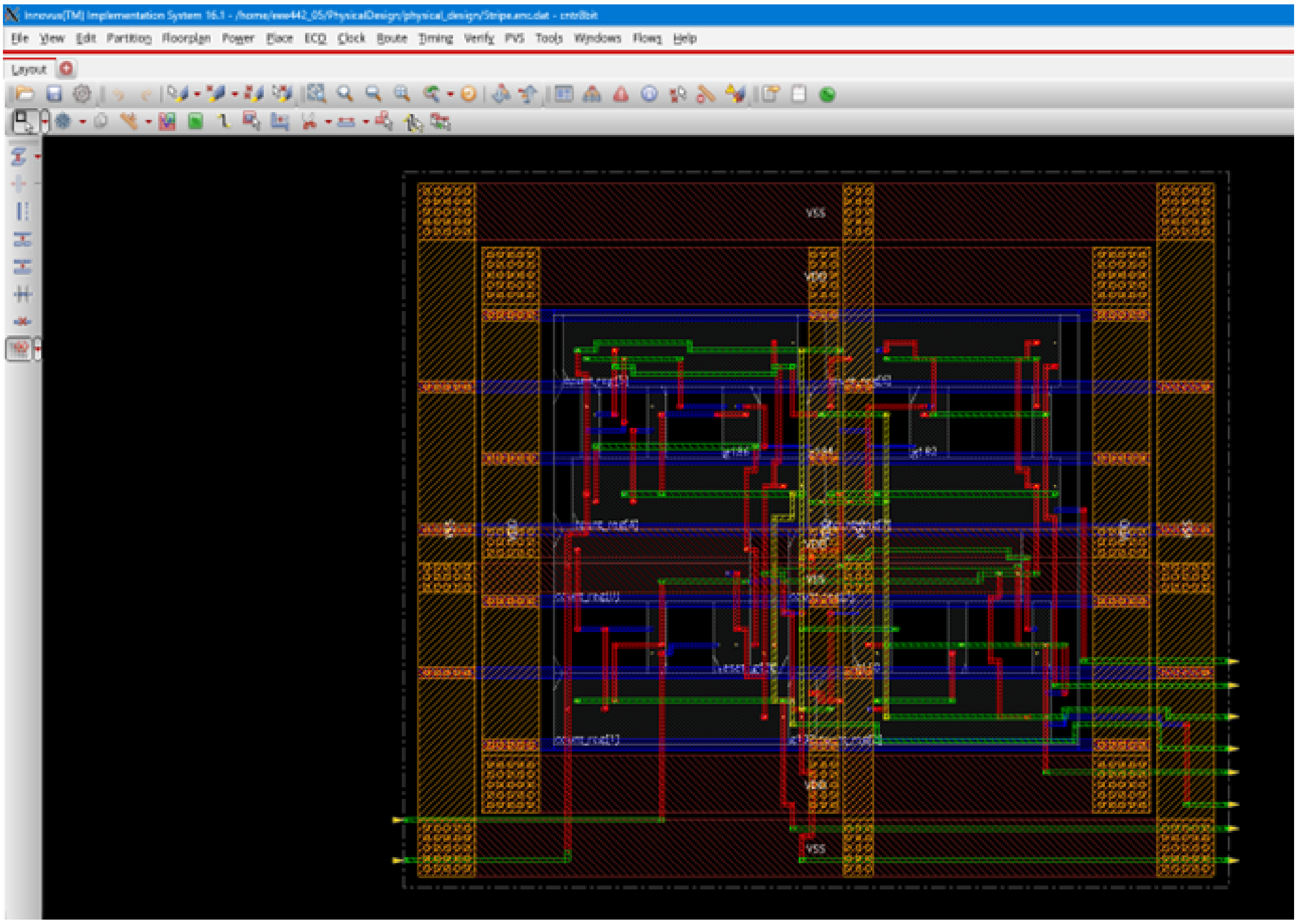

Figure 4.1: Physical Design of 8-bit Counter (Floorplan, Placement, Routing, and Layout)

Chapter 5

Verification

5.1 Geometry Verification

The design was verified for geometry correctness. No violations were found.

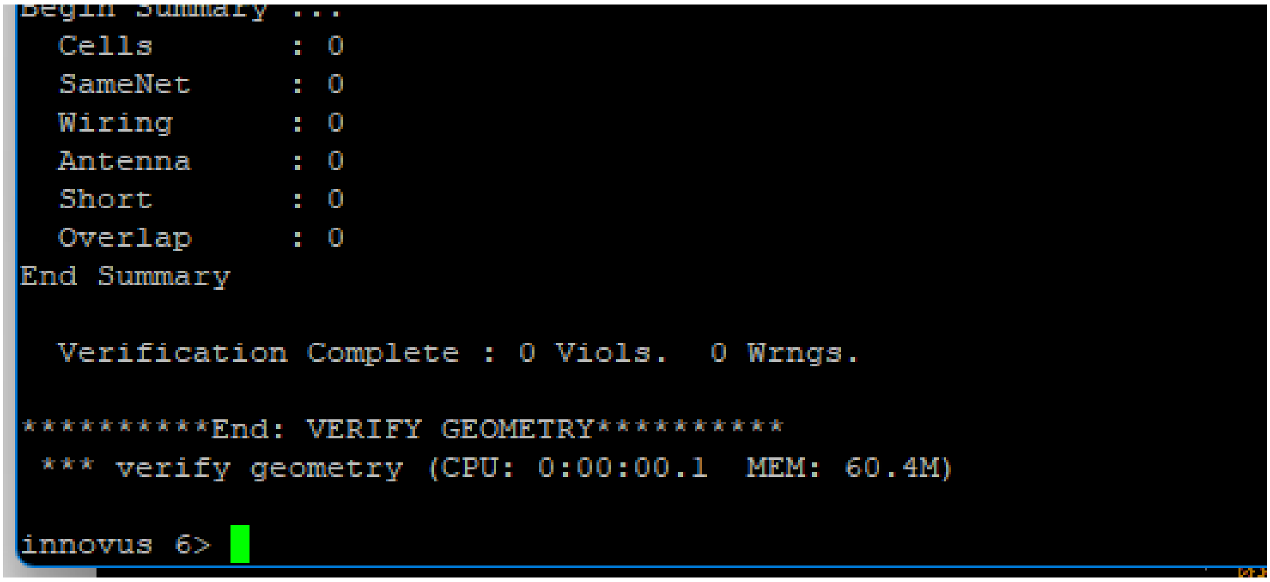

Figure 5.1: Geometry Verification Result (0 Violations)

5.2 Connectivity Verification

All connections were verified for correctness between layout and schematic levels.

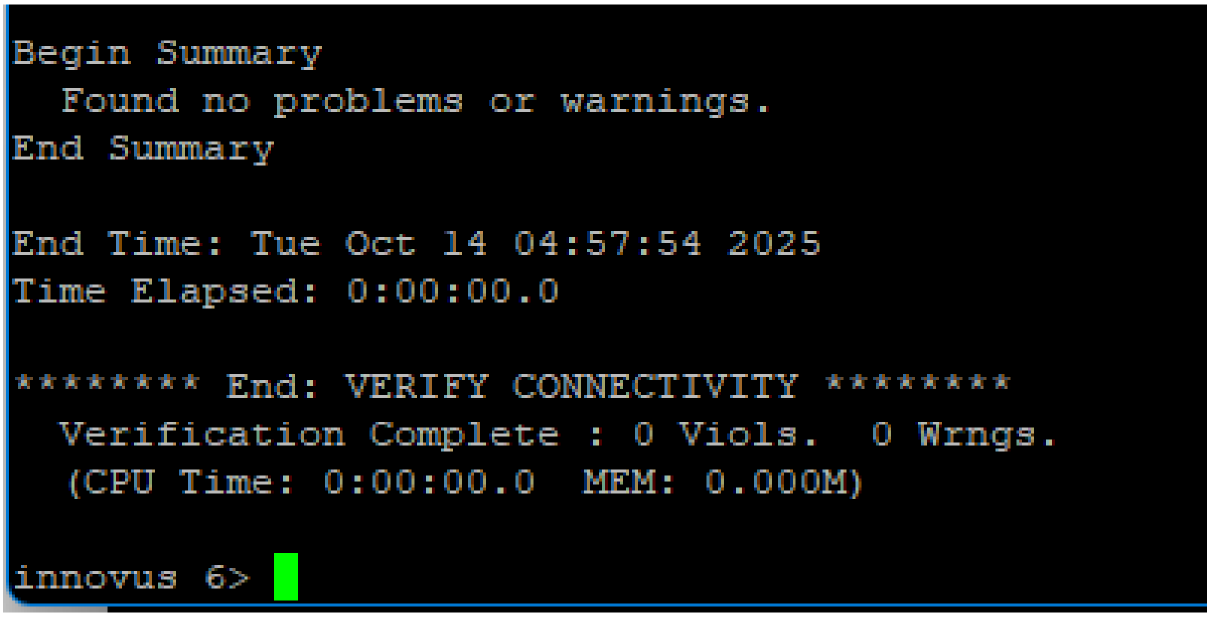

Figure 5.2: Connectivity Verification Result (Clean Connectivity)

5.3 Conclusion

The complete RTL to GDSII implementation of an 8-bit counter was successfully carried out. The design passed all synthesis and verification stages, including geometry and connectivity checks, with zero violations. This confirms that the final layout is fabrication-ready and functionally correct.